

Functions, Linear Equations & *Systems*

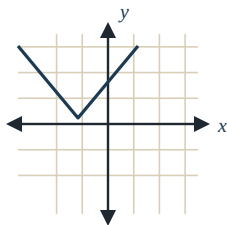
25 problems covering Sections 6.1, 6.2, 6.3, and 6.7. All answers and explanations included.

Quick Reference Unit 6 Key Formulas & Concepts

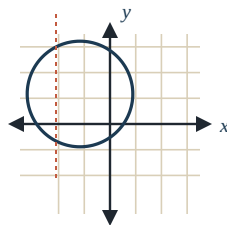
Function (6.1)

A relation in which every **input (x)** has exactly **one output (y)**.

Vertical Line Test: if any vertical line crosses the graph more than once, it's *not* a function.



✓ **Function**



✗ **Not a function**

Function Notation (6.1)

$f(x) = 2x + 4$ reads "*f of x*".

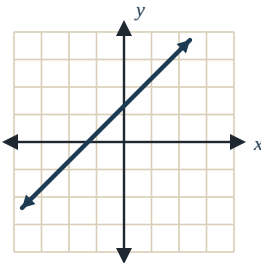
To **evaluate**, substitute the input for x using parentheses.

Example: $f(3) = 2(3) + 4 = 10$.

Domain = all x -values (inputs)

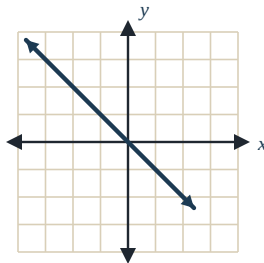
Range = all y -values (outputs)

Four Types of Slope (6.2)



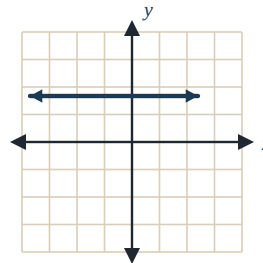
Positive

Up left to right
 $m > 0$



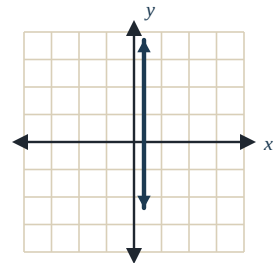
Negative

Down left to right
 $m < 0$



Zero

Horizontal
 $m = 0$



Undefined

Vertical
 $m = \text{undefined}$

Slope-Intercept Form (6.2)

$$y = mx + b$$

m = slope (rate of change; the number multiplied by x)

b = y -intercept (where the line crosses the y -axis; constant)

Slope formula: $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}$

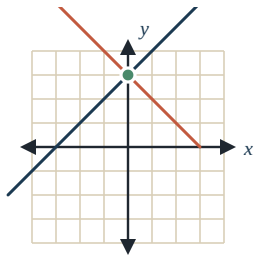
Table → Equation (6.3)

Linear if $\Delta y \div \Delta x$ is the same between every pair of rows.

Steps: (1) Find slope $m = (y_2 - y_1) \div (x_2 - x_1)$. (2) Find b : use the row where $x = 0$, or solve $y = mx + b$ for b . (3) Write $y = mx + b$.

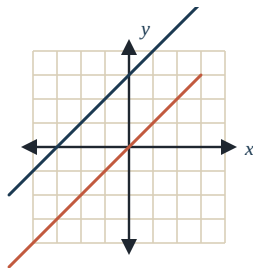
Systems of Equations (6.7)

A **solution** is the (x, y) point that makes *both* equations true — graphically, where the lines intersect.



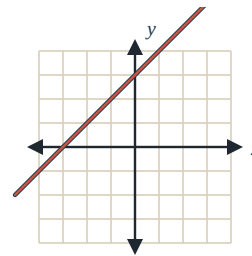
One Solution

Different slopes



No Solution

Same slope, different b



Infinite Solutions

Same slope, same b

Substitution: Solve one equation for a variable, substitute that expression into the other equation, solve, then plug back in to find the second variable.

6.1 Introduction to Functions

1. VOCABULARY

Which statement **best defines** a function?

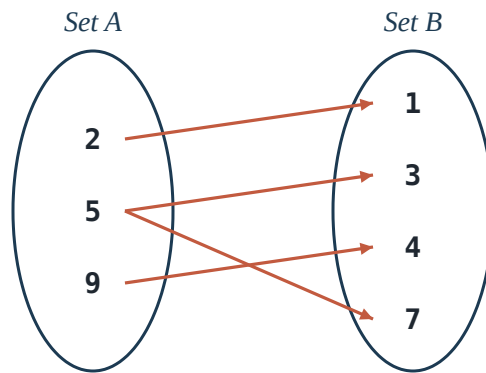
- A. A function is any equation that contains an x and a y .
- B. A function is a relation in which every input has exactly one output. ✓
- C. A function is a relation in which every output has exactly one input.
- D. A function is the graph of a straight line.

ANSWER B

A function pairs each input (x -value) with *exactly one* output (y -value). The same input cannot produce two different outputs.

2. MAPPING DIAGRAM

Does the mapping diagram below represent a function? Why or why not?



- A. Yes — every input is paired with an output.
- B. No — the input 5 is paired with two different outputs (3 and 7). ✓
- C. Yes — no two outputs are repeated.
- D. No — the output 7 is paired with only one input.

ANSWER B

A relation fails the function test when any single input maps to more than one output. Here, $5 \rightarrow 3$ AND $5 \rightarrow 7$, so it is not a function.

3. ORDERED PAIRS

Which set of ordered pairs does *NOT* represent a function?

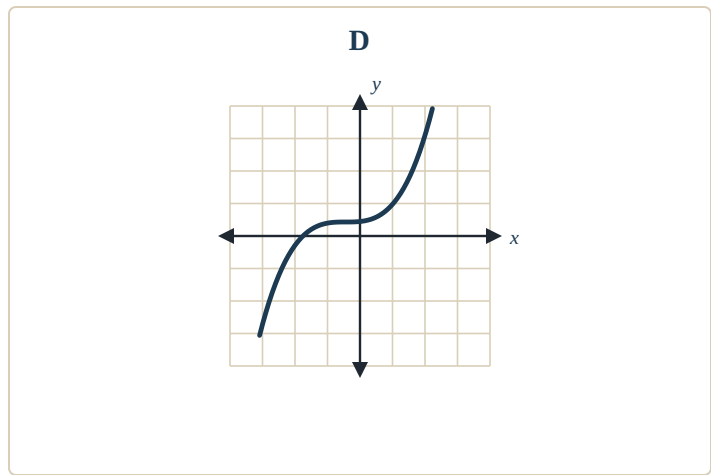
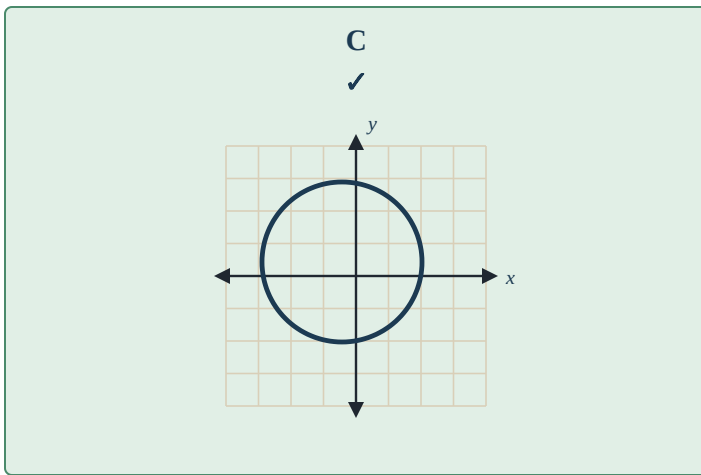
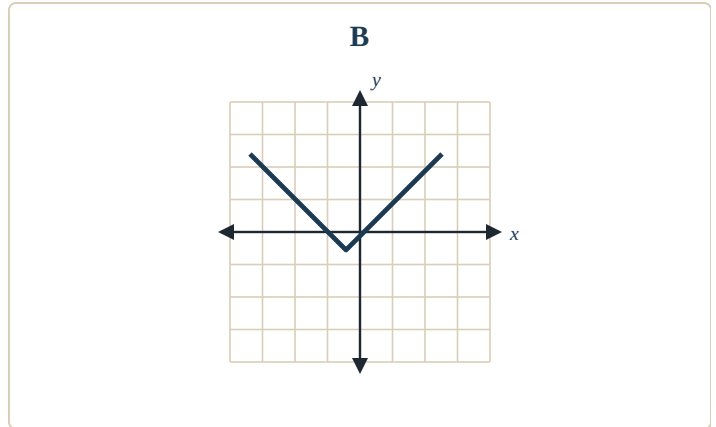
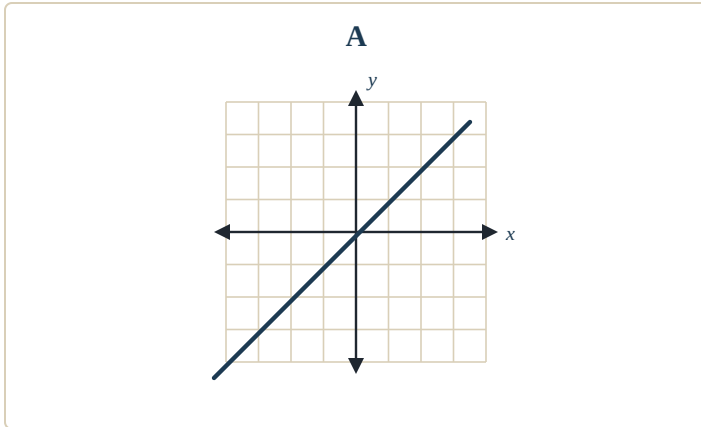
- A. $\{(1, 2), (3, 4), (5, 6), (7, 8)\}$
- B. $\{(-2, 5), (0, 5), (4, 5), (8, 5)\}$
- C. $\{(3, 1), (3, 2), (5, 4), (6, 7)\}$ ✓
- D. $\{(-1, 0), (0, -1), (1, 1), (2, 3)\}$

ANSWER C

Look at the x-values (inputs). In choice C, the input 3 appears twice — paired with 1 and with 2. Choice B is fine: repeated *outputs* are allowed; it's repeated *inputs* with different outputs that aren't.

4. VERTICAL LINE TEST

Which graph does *NOT* represent a function? (Apply the vertical line test.)



ANSWER C

Graph C is a circle. A vertical line drawn through it crosses in two places — so the circle is not a function. Graphs A (line), B (V-shape), and D (cubic) all pass the vertical line test.

5. FUNCTION NOTATION

Given $f(x) = 2x - 7$, find $f(-3)$.

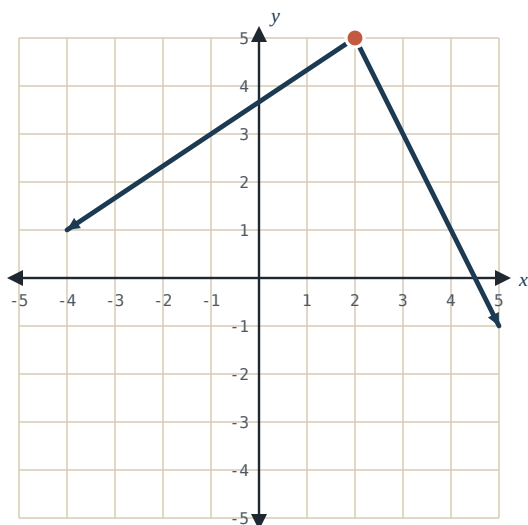
ANSWER $f(-3) = -13$

Substitute -3 for x using parentheses:

$$f(-3) = 2(-3) - 7 = -6 - 7 = -13$$

6. EVALUATE FROM GRAPH

Use the graph of $y = f(x)$ shown below to find $f(2)$.



ANSWER $f(2) = 5$

Find $x = 2$ on the x -axis, then move up (or down) until you hit the graph. The graph passes through $(2, 5)$, so $f(2) = 5$.

7. DOMAIN & RANGE

Find the **domain** and **range** of the function defined by the table below. Express each as a set in $\{ \}$ braces.

x	-3	0	5	8
y	4	-2	7	-2

ANSWER Domain = $\{-3, 0, 5, 8\}$ • Range = $\{-2, 4, 7\}$

The domain is the set of all x -values (inputs); the range is all y -values (outputs). Don't list duplicates: $y = -2$ appears twice in the table but only once in the range.

6.2 Linear Functions & Slope

8. VOCABULARY

For each line description, identify the type of slope.

A horizontal line

Zero

A vertical line

Undefined

A line going up from left to right

Positive

A line going down from left to right

Negative

ANSWER Horizontal $\rightarrow$ Zero • Vertical $\rightarrow$ Undefined • Up $\rightarrow$ Positive • Down $\rightarrow$ Negative

A horizontal line has slope 0 (no rise). A vertical line has undefined slope (no run; you'd divide by 0). Lines that rise as x increases are positive; those that fall are negative.

9. SLOPE FROM TWO POINTS

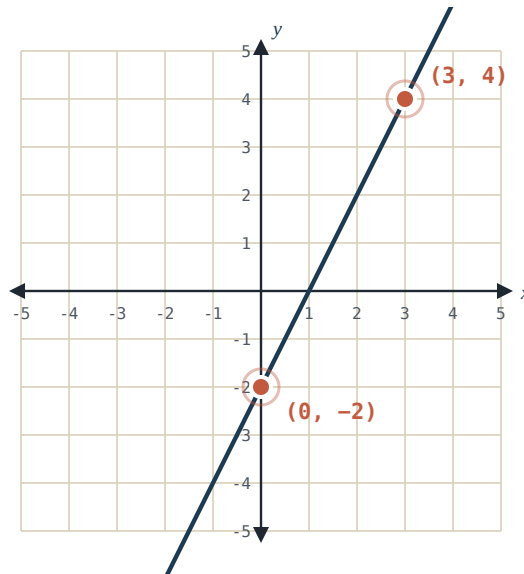
Find the slope of the line through $(-2, 5)$ and $(4, -1)$. Write your answer in simplest form.

ANSWER $m = -1$

$$m = (y_2 - y_1) \div (x_2 - x_1) = (-1 - 5) \div (4 - (-2)) = -6 \div 6 = -1$$

10. SLOPE FROM GRAPH

Find the slope of the line shown on the graph below. Two points on the line are marked for you.



ANSWER $m = 2$

Use the two marked points $(0, -2)$ and $(3, 4)$. Then $m = (4 - (-2)) \div (3 - 0) = 6 \div 3 = 2$.

11. IDENTIFY SLOPE

Find the **slope** of the line: $y = -\frac{3}{4}x + 5$

ANSWER $m = -3/4$

In $y = mx + b$, the slope m is the number multiplied by x . Here, $m = -3/4$.

12. IDENTIFY Y-INTERCEPT

Find the **y-intercept** of the line: $y = 2x - 9$

ANSWER $b = -9$

In $y = mx + b$, the y-intercept b is the constant. Here, $b = -9$ — the line crosses the y-axis at $(0, -9)$.

13. LINEAR OR NOT

Which table represents a **linear function**?

A

x	0	1	2	3
y	1	2	4	8

B

x	-2	-1	0	1
y	4	1	0	1

C

✓

x	0	1	2	3
y	7	4	1	-2

D

x	0	1	2	3
y	0	1	4	9

ANSWER C

Check $\Delta y \div \Delta x$ between consecutive rows. In choice C: $4-7 = -3$, $1-4 = -3$, $-2-1 = -3$, all over $\Delta x = 1$. Constant slope of $-3 \rightarrow$ **linear**. The other tables have changing rates of change.

14. ERROR ANALYSIS

A student tried to find the slope of the line through $(2, -3)$ and $(8, 9)$. Their work is shown below. Identify the error and find the **correct slope**.

STUDENT'S WORK

$$m = (8 - 2) \div (9 - (-3))$$

$$m = 6 \div 12$$

$$m = 1/2$$

ANSWER Correct slope: $m = 2$

The Error: The student switched x and y. The slope formula is $(y_2 - y_1) \div (x_2 - x_1)$, so the y-values go on top, not the x-values. The student got the reciprocal of the right answer.

Correct work: $m = (9 - (-3)) \div (8 - 2) = 12 \div 6 = 2$

15. SOLUTIONS TO A LINEAR EQUATION

Which of the following is a **solution** to the linear equation $y = -2x + 5$?

A. $(1, 2)$

B. $(3, -1)$ ✓

C. $(0, -5)$

D. $(-2, 1)$

ANSWER B

An ordered pair (x, y) is a solution when it makes the equation true. Test $(3, -1)$: $-1 = -2(3) + 5 = -6 + 5 = -1$ ✓. The other points don't satisfy the equation: $(1, 2) \rightarrow -2(1) + 5 = 3 \neq 2$.

6.3 Writing Equations from Tables

16. MISSING VALUE

Assuming a linear relationship, find the **missing value** in the table below. (Use the Δx and Δy columns to find the pattern.)

x	y	Δx	Δy
1	7	—	—
2	11		
3	15		
4	19		
5	?		

ANSWER $y = 23$

Each step adds $\Delta x = +1$ and $\Delta y = +4$ (slope = 4). After $y = 19$, the next y is $19 + 4 = 23$.

17. IDENTIFY LINEAR TABLE

Which table represents a **linear function**?

A

x	1	2	3	4
y	2	6	12	20

B

✓

x	-3	-1	1	3
y	8	4	0	-4

C

x	0	1	2	3
y	5	6	8	11

D

x	0	2	4	6
y	1	3	7	13

ANSWER B

In choice B, x increases by 2 each row and y decreases by 4. Slope = $-4 \div 2 = -2$, the same every time → **linear**. The other tables have changing rates.

18. EQUATION FROM TABLE (X = 0)

Find the equation of the linear function in **slope-intercept form**. (Use the Δx and Δy columns to find the slope.)

x	y	Δx	Δy
0	-5	—	—
1	-2		
2	1		
3	4		
4	7		

ANSWER $y = 3x - 5$

Slope: each row adds $\Delta x = +1$ and $\Delta y = +3$, so **$m = 3$** . Y-intercept: when $x = 0$, $y = -5$, so **$b = -5$** . Equation: **$y = 3x - 5$** .

19. EQUATION FROM TABLE (NO $x = 0$)

Find the equation of the linear function in **slope-intercept form**. Note: the table does not include the $x = 0$ row. (Find slope first, then solve for b .)

x	y	Δx	Δy
1	1	—	—
2	-3		
3	-7		
4	-11		

ANSWER $y = -4x + 5$

Slope: $\Delta x = +1$, $\Delta y = -4$ each row, so **$m = -4$** . To find b , plug slope and $(1, 1)$ into $y = mx + b$: $1 = -4(1) + b \rightarrow b = 5$. Equation: **$y = -4x + 5$** .

20. ERROR ANALYSIS

A student wrote an equation for the linear table below. Identify the error and write the **correct equation** in slope-intercept form.

x	y	Δx	Δy
0	4	—	—
1	7		
2	10		
3	13		

STUDENT'S WORK

Student's answer: $y = 4x + 3$

ANSWER Correct equation: $y = 3x + 4$

The Error: The student switched the slope and the y-intercept. The slope is the rate of change ($\Delta y = +3$ per $\Delta x = +1$, so $m = 3$), and the y-intercept is the y-value when $x = 0$ (which is 4, so $b = 4$).

$\Delta y \div \Delta x = 3 \div 1 = 3$; $b = 4$ (the y-value when $x = 0$). Correct: $y = 3x + 4$.

6.7 Systems of Equations

21. VOCABULARY

A **solution** to a system of two linear equations is best described as...

- A. any point that lies on either line.
- B. the point where both lines have the same slope.
- C. the **(x, y) coordinate that makes both equations true.** ✓
- D. the x-value where one line equals zero.

ANSWER C

A solution must satisfy both equations at once. On a graph, that's the point where the two lines intersect — the same (x, y) works in both.

22. SOLUTIONS BY INSPECTION

How many solutions does this system have? $y = 2x + 5$ $y = 2x - 1$

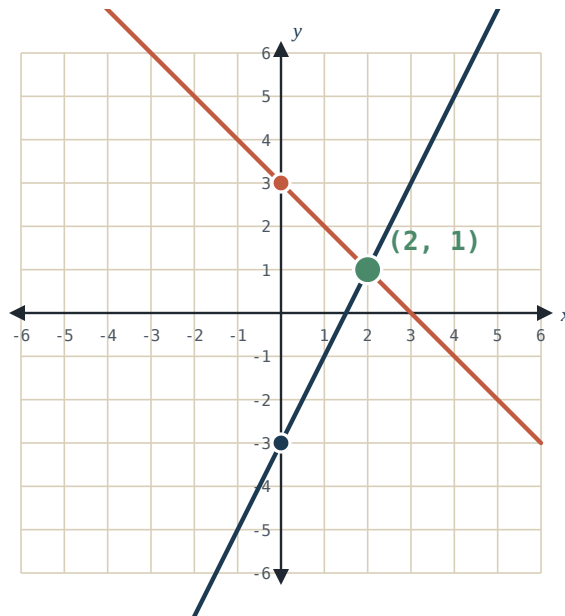
- A. One solution
- B. **No solution** ✓
- C. Infinite solutions
- D. Cannot be determined

ANSWER B

Both lines have slope 2 (same slope) but different y-intercepts (5 and -1). They are **parallel**, so they never intersect → **no solution**.

23. SOLVE BY GRAPHING

Graph the following system on the grid below and find the solution. (Each line's y-intercept appears on the grid — start there, then use the slope.)



$$y = 2x - 3$$

$$y = -x + 3$$

ANSWER Solution: (2, 1)

Graph both lines. The first ($y = 2x - 3$) has y-intercept -3 and slope 2 ; the second ($y = -x + 3$) has y-intercept 3 and slope -1 . They cross at **(2, 1)**. Check: $1 = 2(2) - 3 = 1$ ✓ and $1 = -(2) + 3 = 1$ ✓.

24. SUBSTITUTION

Solve the system by substitution. $y = 2x + 1$ $y = -x + 7$

ANSWER Solution: (2, 5)

Both equations equal y , so set them equal:

$$2x + 1 = -x + 7 \rightarrow 3x = 6 \rightarrow x = 2$$

Substitute into either equation: $y = 2(2) + 1 = 5$. Solution: **(2, 5)**.

25. SUBSTITUTION

Solve the system by substitution. $y = x - 4$ $3x + 2y = 7$

ANSWER Solution: (3, -1)

Substitute $(x - 4)$ for y in the second equation:

$$3x + 2(x - 4) = 7 \rightarrow 3x + 2x - 8 = 7 \rightarrow 5x = 15 \rightarrow x = 3$$

Then $y = 3 - 4 = -1$. Solution: **(3, -1)**.